

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats

# Create data
age = [23, 23, 27, 27, 39, 41, 47, 49, 50,
       52, 54, 54, 56, 57, 58, 58, 60, 61]

fat = [9.5, 26.5, 7.8, 17.8, 31.4, 25.9, 27.4, 27.2, 31.2,
       34.6, 42.5, 28.8, 33.4, 30.2, 34.1, 32.9, 41.2, 35.7]

df = pd.DataFrame({"Age": age, "%Fat": fat})

# Mean, Median and Standard Deviation
print("Mean:\n", df.mean())
print("\nMedian:\n", df.median())
print("\nStandard Deviation:\n", df.std())

# Boxplots
df.boxplot(column=["Age", "%Fat"])
plt.title("Boxplot of Age and Body Fat")
plt.show()

# Scatter plot
plt.scatter(df["Age"], df["%Fat"])
plt.xlabel("Age")
plt.ylabel("Body Fat (%)")
plt.title("Age vs Body Fat")
plt.show()

# Q-Q plots
stats.probplot(df["Age"], dist="norm", plot=plt)
plt.title("Q-Q Plot of Age")
plt.show()

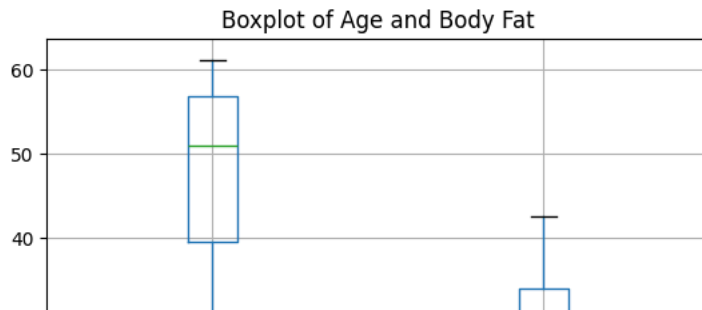
stats.probplot(df["%Fat"], dist="norm", plot=plt)
plt.title("Q-Q Plot of Body Fat")
plt.show()
```



```
Mean:
Age      46.444444
%Fat     28.783333
dtype: float64
```

```
Median:
Age      51.0
%Fat     30.7
dtype: float64
```

```
Standard Deviation:
Age      13.218624
%Fat     9.254395
dtype: float64
```



```
import pandas as pd
from google.colab import files

# Select and upload CSV file
uploaded = files.upload()

# Read the uploaded CSV file
df = pd.read_csv(next(iter(uploaded)))

# Calculate Total Sales
df["Total Sales"] = df["Quantity Sold"] * df["Unit Price"]

# Product-wise total sales
product_sales = df.groupby("Product")["Total Sales"].sum()

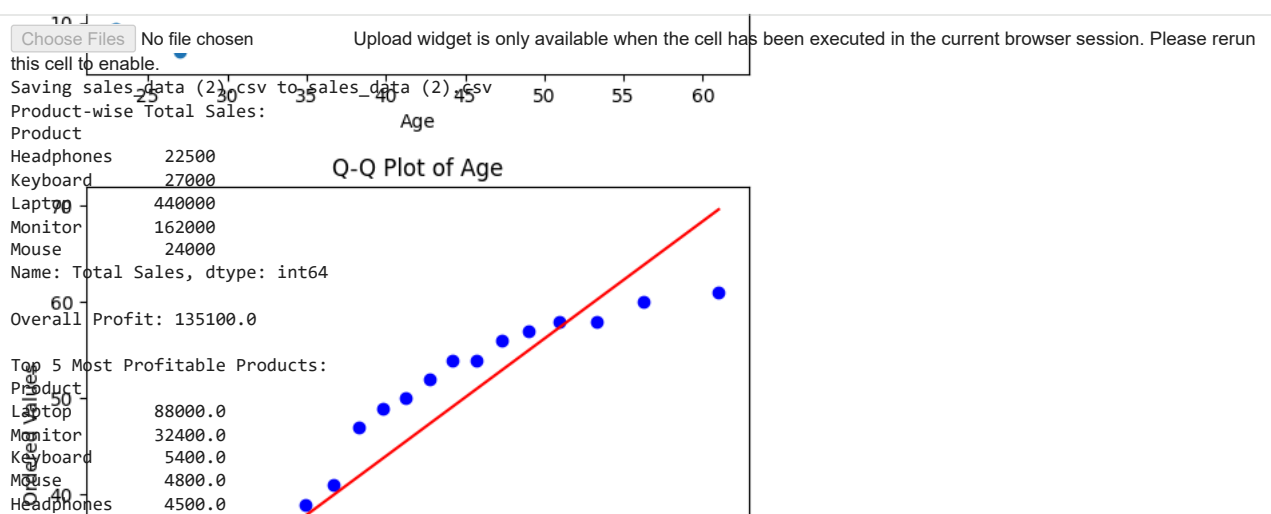
# Calculate profit
product_profit = product_sales * 0.20

# Overall profit
overall_profit = product_profit.sum()

print("Product-wise Total Sales:")
print(product_sales)

print("\nOverall Profit:", overall_profit)

print("\nTop 5 Most Profitable Products:")
print(product_profit.sort_values(ascending=False).head(5))
```



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